Easy Geometrical Problems — problem 12.3.1.33

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Let P be any point inside a triangle with distances l_a, l_b, l_c from the sides a, b, c. We may assume $a \leq b \leq c$. Show that $h_c \leq l_a + l_b + l_c \leq h_a$. There is equality iff the triangle is regular.